Immunet, which uses ClamAV, is available as a cloud based antivirus. The most interesting feature of Immunet is that it requires no updates of virus definitions on your system. Though Immunet was acquired by *sourcefire.com*, its free edition is available on *http://www.immunet.com/free/index.html*.

Though not an open source product, Zonealarm is a free antivirus with a firewall and is worth considering if you are an individual home user or need it for a certain type of not-for-profit organisation. This product from Checkpoint is simple to install and a very effective antivirus. Further, you can configure its firewall to allow only certain programs from your Windows system to access the Internet. Look at it this way - if you do not wish to allow particular software to access the Internet (like CD-burning software, installers of various software, etc), use a firewall to block them.

Nmap

I believe any article on LAN tools must feature Nmap, which is the first tool that comes to mind for day-to-day network admin tasks.

As mentioned on *nmap.org*, Nmap (Network Mapper) is a free and open source (licence) utility for network discovery and security auditing.

Typically, Nmap discovery leads to the answer of the question, 'What constitutes my network?' A typical process to use Nmap for discovery could start with scanning the IP range for active IP addresses. The scan technique to be used depends on your privilege level. For a non-administrator, a ping scan or a TCP connect scan could be used, whereas an administrator (root) can use a Syn scan. Once the active IP addresses are identified, enumeration of open ports (services) could be fired and finally, you may wish to identify the service running on the open port since often, non-standard ports are used to run services. For example, many a time, the Apache server default http port (80) is changed to some other port.

A few simple things that you can discover by using Nmap are:

- Active IP addresses in your LAN
- Open ports and corresponding running services
- Detecting the operating system on IP addresses

You could study *http://nmap.org/nsedoc/categories/discovery.html* for all readymade scripts and their uses.

Wireshark: Can you diagnose network problems without it?

Troubleshooting LAN problems is like blind firing or a process of trial and error, till you locate the problem. Wireshark is an excellent tool, which can read the 'traffic on wire' – basically, all network traffic originating from the system on which it is installed. Surprisingly, it is free and available for Linux as well as Windows.

Packets captured using Wireshark can be viewed from perspectives of TCP/IP layers:

1. Frame - the overall data
2. Ethernet II - MAC addresses
3. Internet Protocol - IP addresses
4. Transmission Control Protocols - source and destination ports and acknowledge numbers.

By looking at captured traffic, you can often locate important troubleshooting information such as packet loss, too much broadcast, the presence of IPv6 in the network even if it is not being used, worm or malware-generating traffic, etc.

There are various Wireshark features that drill through the captured traffic such as:

- Analysis - Display filter, decode as, follow TCP stream
- Statistics - Summary, protocol hierarchy

Spiceworks

Only administrators will know the difficulty of preparing a list of IT assets without an automated tool. Managements do not consider investing in inventory tools due to budgetary constraints. Spiceworks offers one of the best solutions, at no cost.

Spiceworks started primarily as a network inventory tool, and has expanded to cover three additional and useful areas:

- Network monitoring
- Helpdesk
- Mobile device management

Spiceworks is really easy to set up, can track hardware and software inventory, and can integrate with the Windows domain. Over and above this, it has a vibrant community ready to help! Go ahead and try it out since it is really simple.

Table 1: A few other useful LAN tools

Tool	Description
Dig	Finds DNS records
Traceroute	Traces the path to the destination IP address
Netstat	Checks open ports on systems
Netcat	Reads and writes data across network connections; it is also termed as the 'Swiss Army knife' for network administrators
Aircrack	Suite of tools for wireless monitoring and security testing
K3B	Linux based CD and DVD writer, which also includes a ripper

mRemoteNG

Most administrators love good old PuTTY. However, for managing various remote connections in addition to SSH, rlogin and Telnet, you may consider mRemoteNG. It also provides support for RDP, VNC and ICA (from Citrix).

One more embedded advantage is the 'copy and paste' feature, which eliminates the requirement of additional software such as WinSCP.

Six other LAN tools are listed in Table 1. The list, however, is endless. END

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